NANYANG
THE INNOVATIVE POLYTECHNIC

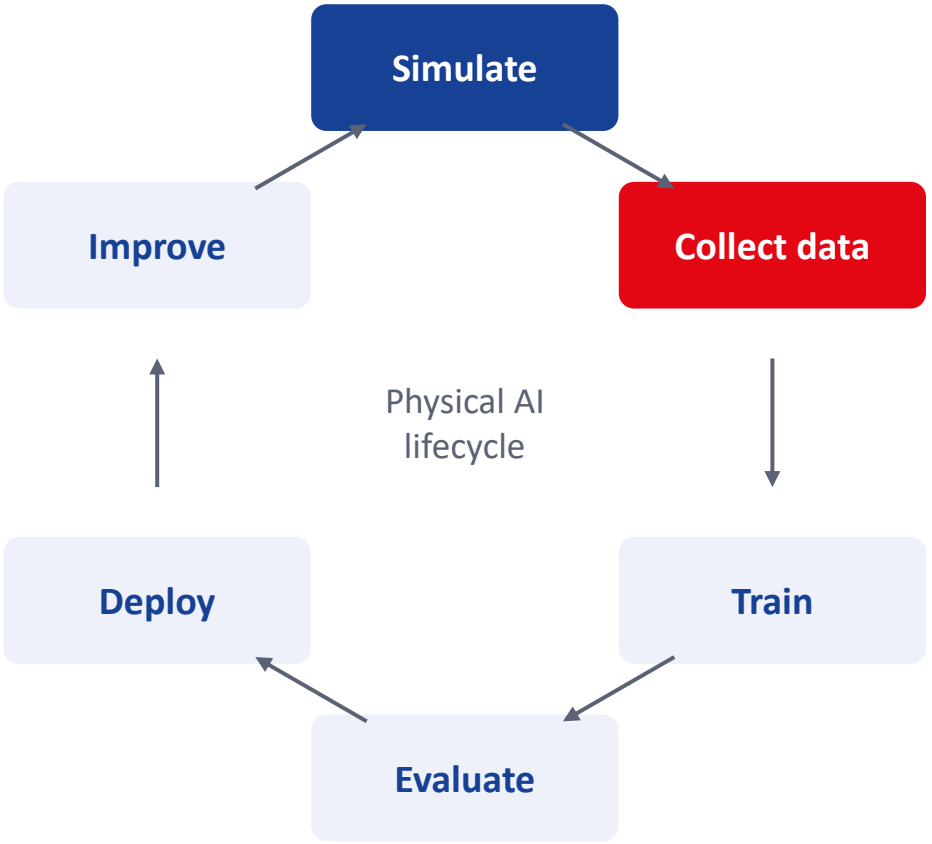
Robot Data

Teleoperation, kinesthetic teaching and simulation

Data Acquisition & Policy Learning · Lesson 3



Where we are: Block B · collecting robot data



A	Foundations	L1–L2
B	Collecting robot data	L3–L4
C	RL in simulation	L5–L6
D	RL: sim → real arm	L7–L8
E	Imitation learning	L9–L10
F	Evaluate & deploy IL	L11–L12
G	Vision-language-action	L13–L14

Entry check: three questions on Lab 2

1. What did calibration establish between the leader and the follower?
2. During teleoperation, what exactly does the follower copy from the leader?
3. At 30 Hz, how many joint values does the SO-101 report per second?

5 minutes, on paper or in the poll. We go through the answers together.

Learning outcomes

- By the end of this lesson you should be able to:
  - describe the structure of a robot dataset: episode, frame, observation, action, task
  - compare teleoperation, kinesthetic teaching and domain-randomised simulation on cost, realism and scale
  - apply quality and coverage rules to decide which demonstrations to keep
  - explain domain randomisation as a response to three kinds of reality gap

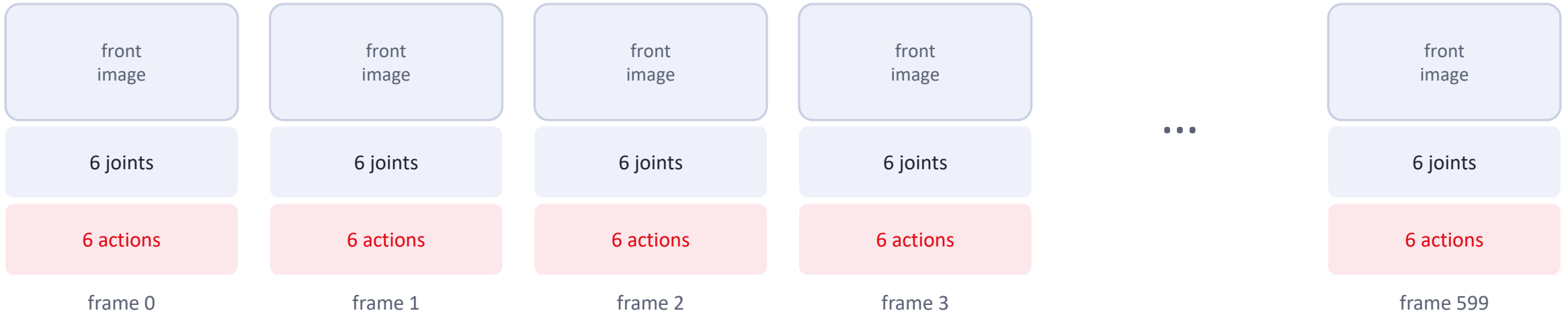
Part 1

What a robot dataset is

Episodes, frames, state and action

A robot dataset is a time series

One episode = one attempt at the task, from reset to finish (here 20 s × 30 fps = 600 frames)



Frame
one time step: observation + action + timestamp

Episode
an ordered sequence of frames for one attempt

Dataset
many episodes of the same task, plus metadata

Inside one frame

Field	Shape	Meaning
observation.state	6 numbers	follower joint positions
observation.images.front	480 × 640 × 3	front camera image
observation.images.wrist	480 × 640 × 3	wrist camera image
action	6 numbers	leader joint positions (the command)
timestamp · frame_index · episode_index	numbers	where this frame sits in time
task	text	“Pick the cube and place it in the bin”

Field names follow LeRobot's dataset format. Camera names (front, wrist) are chosen by you when recording and must stay the same.

State vs action: command and result



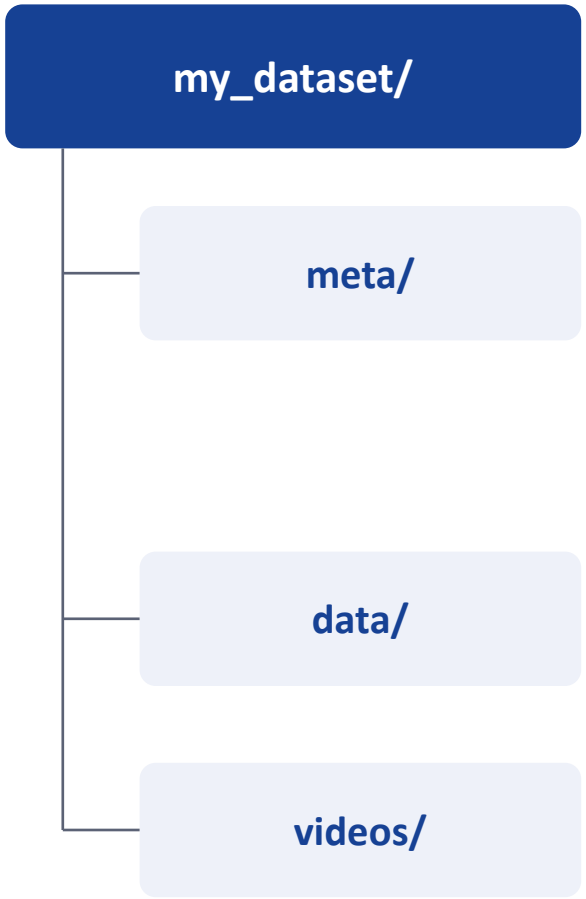
Illustrative curve. Replace with a joint trace from your own L2 teleoperation.

Action
where the follower was told to go

State
where the follower actually is

The follower lags a few frames and may not reach the command (for example when gripping). The policy learns to output commands.

How LeRobot stores a dataset (format v3.0)



info.json · schema, fps, features
stats.json · normalisation statistics
tasks · the task text
episodes/ · lengths and offsets

Parquet files · numbers for every frame (state, action, timestamps, indices)

MP4 files · one stream per camera (front, wrist)

Why this design?

Numbers are small and fast to read: Parquet.
Images are huge: stored as compressed video.
Many episodes per file: quick to load and stream from the Hugging Face Hub.

Normalisation: the same scaling at training and on the arm

During training
Each signal is rescaled using its mean and spread from stats.json, so the network sees numbers of similar size.

On the real arm
The same rescaling must be applied to live readings, and undone on the policy's output. LeRobot does this for you.

Example: elbow_flex mean = 60, spread = 20. A reading of 80 becomes (80 − 60) ÷ 20 = **+1.0**
If the scaling were different on the arm, the policy would see a different number for the same pose.

Worked example: how big is our dataset?

50 episodes × 20 seconds × 30 frames per second

30,000
frames

60,000
camera images (2 cameras)

180,000
joint values in the state

≈ 1 hour
to record, with resets

Compare: ImageNet has over a million images, collected by scraping. Every one of our 60,000 images comes from a person physically performing the task.

Part 2

Three ways to collect

Teleoperation, kinesthetic teaching and domain-randomised simulation

Three ways to collect robot data

Teleoperation

Control the robot through a separate device

- keyboard, gamepad, SpaceMouse, VR, or our SO-101 leader arm
- you never touch the robot that does the task
- the leader arm needs no retargeting: its joints match the follower's 1:1

Kinesthetic teaching

Physically guide the robot itself

- motors off or in a compliant mode; you push the arm through the task
- the robot records its own joint positions
- e.g. Franka guiding mode, UR Freedrive, SO-101 follower with torque off

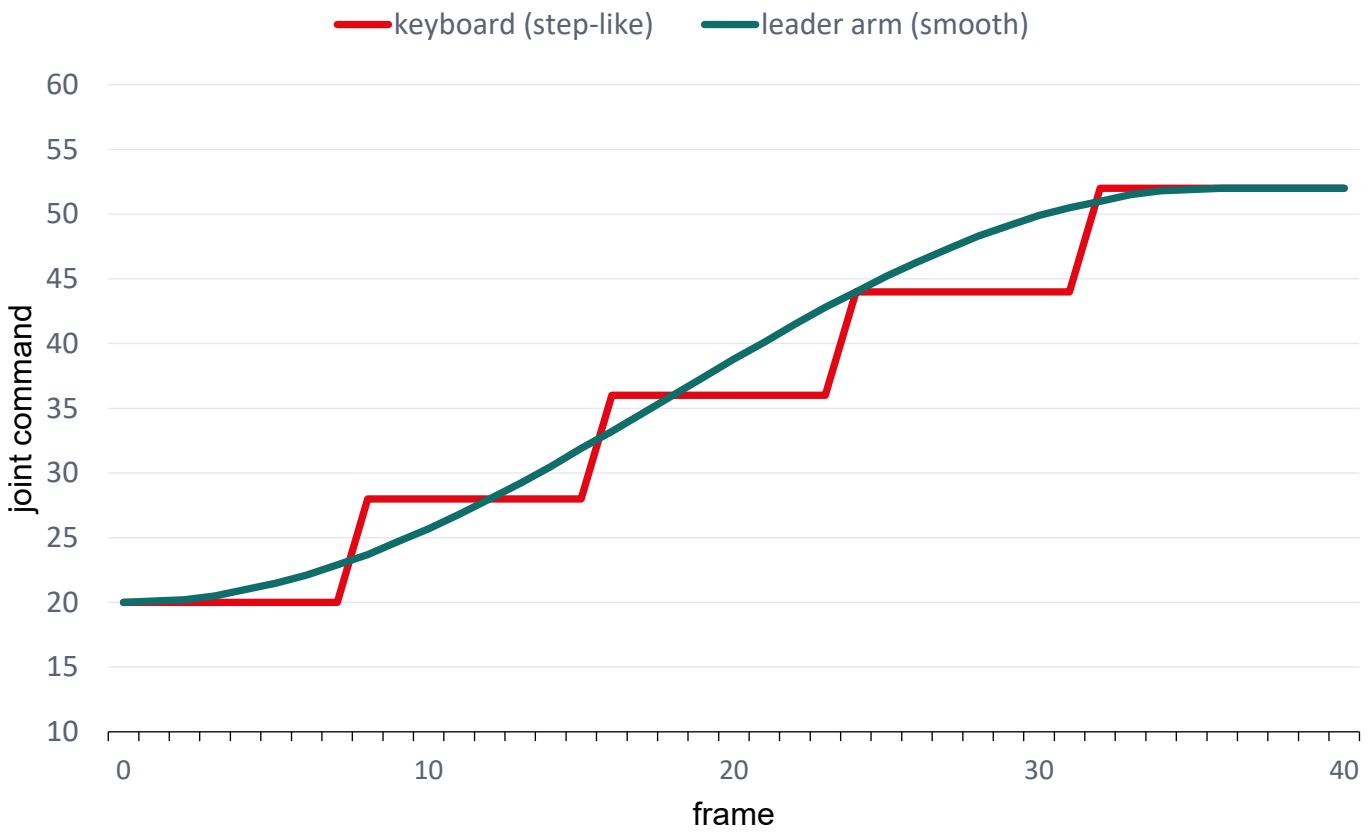
Domain-randomised simulation

Generate data in a varied virtual world

- physics, looks and sensor noise change every episode
- cheap, scalable, labels for free

The test: does your hand touch the robot that does the task? No → teleoperation. Yes → kinesthetic teaching. Leader–follower is teleoperation, as in LeRobot, ALOHA and GELLO.

Teleoperation: the device leaves a signature



Illustrative traces of the same reach: keyboard teleop vs leader-arm teleop.

What a device does to the data

- Keyboard: fixed-size steps, pauses between key presses
- Gamepad / SpaceMouse: smoother, but still retargeted
- VR hand tracking: natural, needs a headset and fast network
- SO-101 leader arm: smooth, joint-matched, no retargeting

The policy copies whatever style the data has.

Kinesthetic teaching: guiding the robot itself

PHOTO / VIDEO
Photo: a hand guiding the SO-101 follower with its torque off

Hand-guiding (lead-through)
torque off or compliant; push the robot itself and it records its joints

Record, then replay
replay the saved path with nobody in view to capture clean camera images

Strengths
direct feel of contact; no retargeting; no second device

Limits
your hand is in the camera images; tiring; you hold the arm's full weight

Worked example: which method does our IL lab use?

Our IL lab (Lab 10): leader–follower

You move the leader
your hand is here

→

lerobot-record
reads both arms, 30 fps

→

Follower does the task
nobody touches it

One recorded frame at t = 4.00 s

action shoulder_lift = 42.0° *leader: the command*

state shoulder_lift = 39.5° *follower: the result*

images front + wrist: arm and cube, no hand

Command and result are both measured, so every frame has a real action.

Hand on the follower? No → **teleoperation**

For contrast: kinesthetic on the SO-101

You push the follower
torque off; your hand is here

→

Follower does the task
moved by your hand

→

It records itself
its own joint positions

One recorded frame at t = 4.00 s

state shoulder_lift = 39.5° *follower*

action none measured: *use the next frame, 40.3° at t + 1/30 s*

images your hand and arm are in view

lerobot-record needs a teleop device, so LeRobot has no built-in recorder for this.

Hand on the follower? Yes → **kinesthetic teaching**

Touching an arm is not the test; touching the arm that does the task is. In the lab you hold the leader, so our IL dataset is teleoperation data.

Check yourself: name the collection method

Set-up	Method	Why
1 Lab 10: you move the SO-101 leader and the follower picks up the cube	**Teleoperation (leader–follower)**	Your hand is on the leader, never on the follower. Action = leader joints.
2 The follower's torque is off and you push it to the cube while it records	**Kinesthetic teaching**	Your hand is on the robot doing the task, and in every camera image.
3 Lab 4 sim station: keyboard keys move the simulated arm	**Teleoperation (in simulation)**	A separate device; fixed key steps give step-like actions.
4 LeIsaac: the real leader arm drives a simulated SO-101	**Teleoperation (into simulation)**	The simulated arm does the task; you only touch the leader.
5 A Franka cobot: press the wrist button and guide the arm to the part	**Kinesthetic teaching**	You hold the robot itself (lead-through).
6 Isaac Lab runs 4,096 arms with random cube mass, friction and colours	**Domain-randomised simulation**	No person per episode; the world changes every time.

Domain-randomised simulation: many worlds, not one

PHOTO / VIDEO
Contact sheet: 16 randomised scenes from Isaac Lab

Physics: cube mass, table friction, motor stiffness

Appearance: colours, textures, lighting

Sensing: noise on joint readings, small offsets

Start state: where the cube is placed each episode

Idea from Tobin et al. (2017): make reality look like one more variation.

Comparing the three methods

	Teleop (incl. leader arm)	Kinesthetic (hand-guiding)	Randomised simulation
Cost per usable episode	medium (human time)	medium–high (human time)	very low once set up
Realism	real robot; style set by device	real robot; hand in camera view	simulated: reality gap
Scale	limited by people	limited by people	thousands in parallel
Who uses it in this course	Lab 4 (keyboard), L9–L14 (leader)	awareness; record-then-replay	RL training (L5–L8)

No method wins on everything. Real projects mix them.

Scaling up (for awareness)

Isaac Lab Mimic

Record about 10 human demos, split them into sub-tasks, and let the simulator re-stitch them in new scenes.

10 demos → 1,000 demos

LeIsaac

The real SO-101 leader arm drives a simulated SO-101, and the demos are recorded straight into LeRobot format.

leader-arm teleop + simulation

Mimic is mentioned only (Linux, GPU-heavy, not needed for our task). LeIsaac appears in Labs 4 and 10 where the lab machines support it.

Part 3

Quality, coverage and randomisation

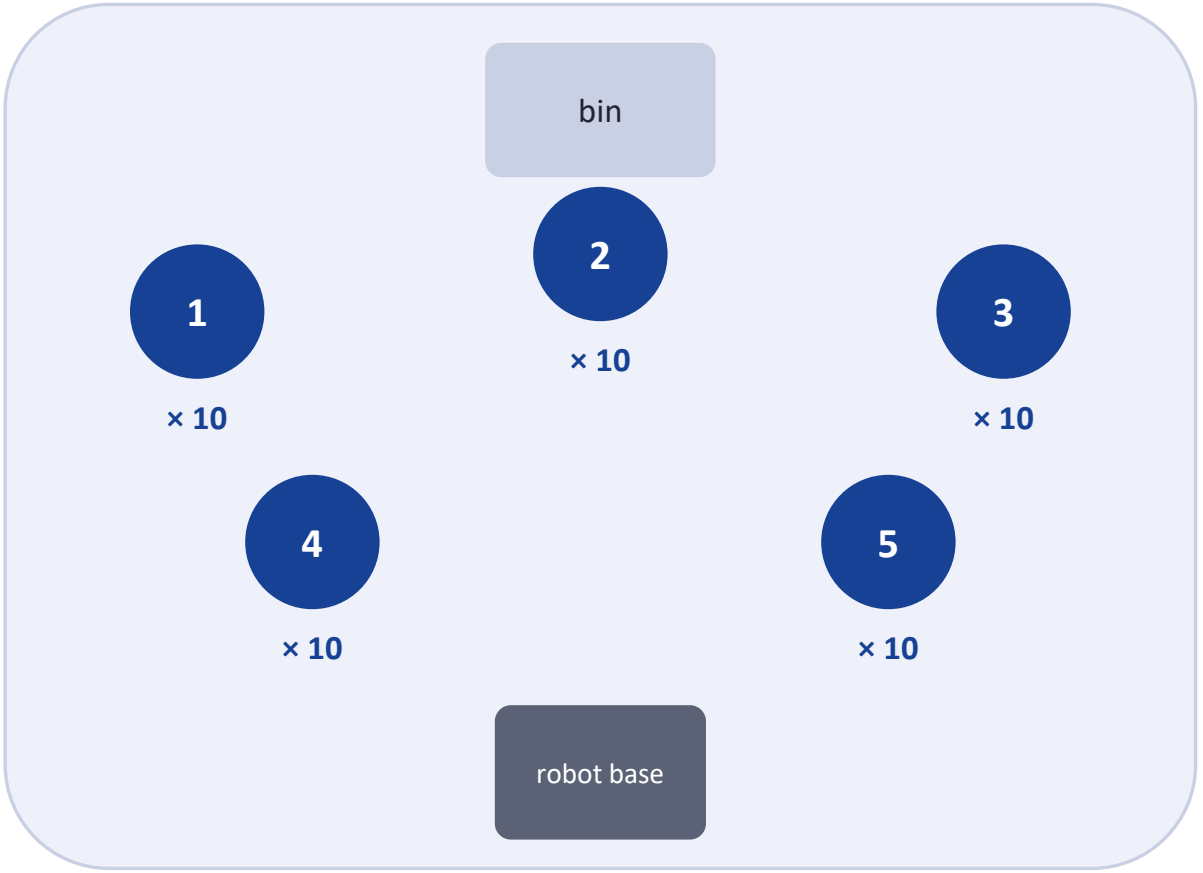
Deciding what to keep, and how wide to vary the world

Good demo, bad demo

Keep	Discard
• smooth and direct • one consistent grasp strategy • no pauses or hesitation • ends cleanly once the cube is in the bin	• pauses or long hesitation • regrasping or a dropped cube • a different strategy from the rest • cut off before the task finished

The policy copies everything, including the pauses and the regrasps. A bad demo is not neutral: it teaches the wrong behaviour.

Coverage: spread the demos over where the policy will work



Grid mat (taped down, never moved)

5 positions × 10 episodes = 50

25 episodes has been shown to be too few
(LeRobot's guidance for SO-100/101 tasks)

Fifty repetitions of one easy position teach the policy one position.

Curation: a written rule, not a feeling

Discard log (one row per episode)

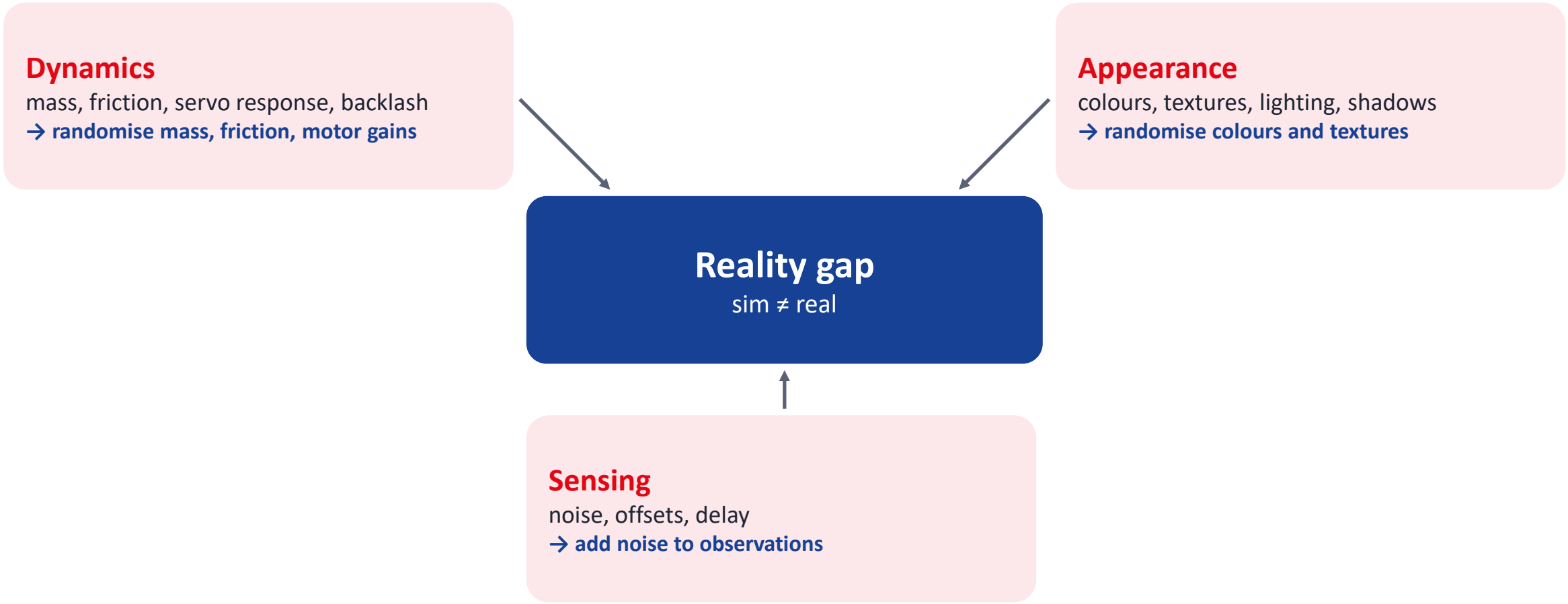
Ep	Keep?	Rule	Note
0	keep	–	
1	discard	pause > 1 s	operator distracted
2	keep	–	
3	discard	regrasp	cube slipped

Datasheet (ships with the dataset)

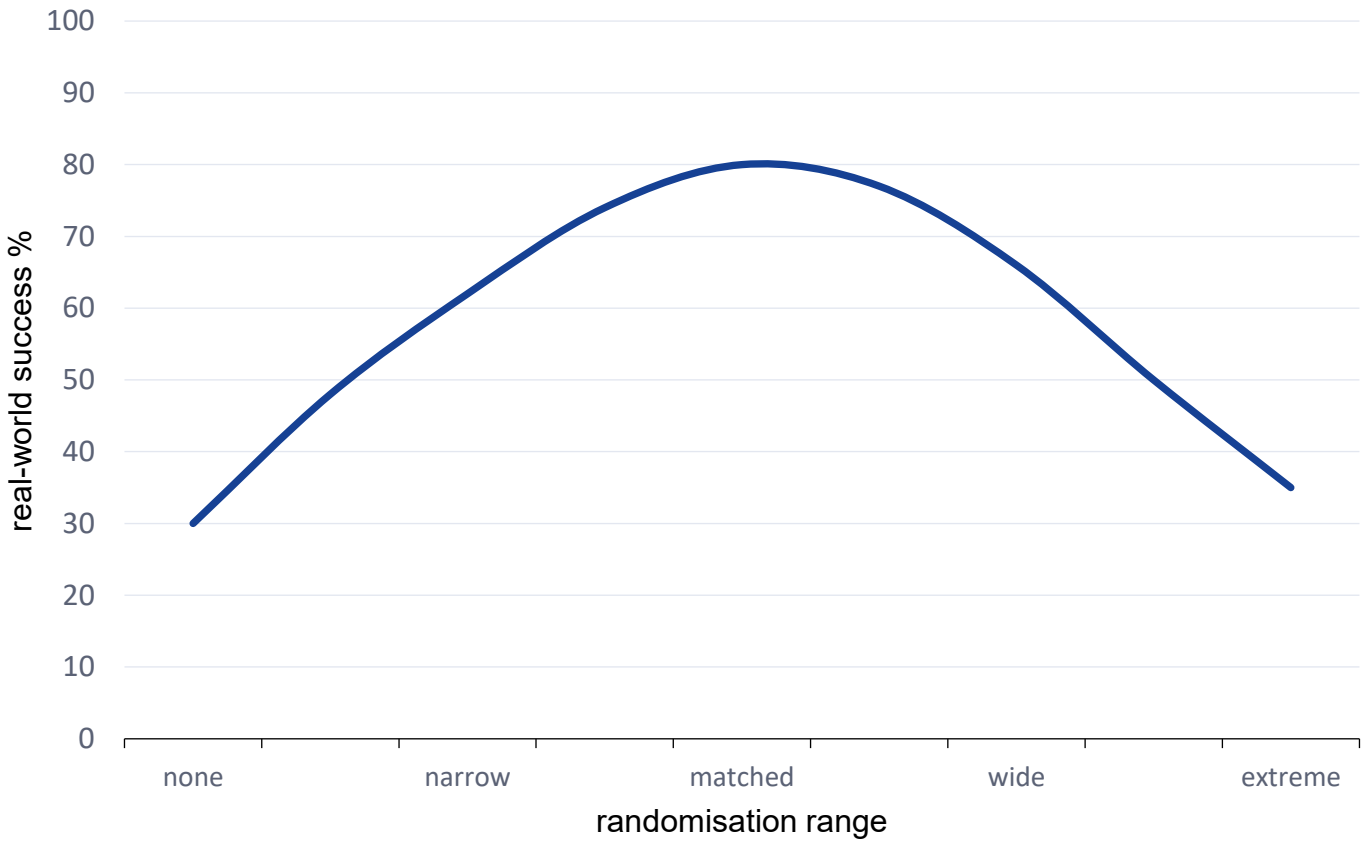
- task text and episode count
- coverage map (which positions, how many)
- camera set-up and names
- known defects
- the discard log

Based on “Datasheets for Datasets” (Gebru et al., 2018). Discarding is normal; hiding what you discarded is not.

The reality gap in three parts



How wide should the randomisation be?



Sketch of the trade-off; not measured data.

Too narrow
the policy overfits the simulator

Too wide
the task becomes hard to learn at all

Matched
cover what you cannot measure, sized to what you can (e.g. measured friction ± a margin)

Activity: Good demo, bad demo

Part A · six clips (10 min)

Watch each clip from the real arm.
Keep or discard? Name the rule.

Clip 1	Clip 2	Clip 3
Clip 4	Clip 5	Clip 6

Part B · choose a method (5 min)

1. Train a policy to reach targets using thousands of attempts.
2. Teach the arm a smooth pick-and-place from 50 examples.
3. Test whether a policy copes with different table colours before using the real arm.

Teleoperation, kinesthetic or randomised simulation? Why?

Exit ticket

- Answer before you leave:
  - A dataset has 40 episodes of 15 s at 30 fps with 2 cameras. How many frames and images?
  - Match each to the gap it closes: friction randomisation · colour randomisation · joint noise
  - State the coverage rule for our pick-and-place dataset.
  - What is the difference between state and action on the SO-101?

Next week: L4 lab · Collect data three ways

Sim station · Isaac Lab

Switch on randomisation; make a 16-scene contact sheet
Name the gap each term targets
Teleoperate the simulated arm and record 5 demos

Arm station · SO-101

Record 10 episodes with the leader arm
Replay one episode on the follower
Inspect the dataset and fill the quality log

Bring: your pod's calibration ids from Lab 2 (the same ids must be used for every recording).

References and further reading

- Hugging Face. LeRobotDataset v3.0. https://huggingface.co/docs/lerobot/lerobot-dataset-v3
- Hugging Face. Imitation Learning on Real-World Robots (recording tips). https://huggingface.co/docs/lerobot/il_robots
- Hugging Face blog. LeRobot Community Datasets. https://huggingface.co/blog/lerobot-datasets
- Tobin et al. (2017). Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World. https://arxiv.org/abs/1703.06907
- OpenAI (2018). Learning Dexterous In-Hand Manipulation. https://arxiv.org/abs/1808.00177
- Weng, L. (2019). Domain Randomization for Sim2Real Transfer. https://lilianweng.github.io/posts/2019-05-05-domain-randomization/
- Gebru et al. (2018). Datasheets for Datasets. https://arxiv.org/abs/1803.09010
- Isaac Lab. Teleoperation and imitation learning. https://isaac-sim.github.io/IsaacLab/v2.3.2/source/overview/imitation-learning/teleop_imitation.html

Advanced topic

Collecting data for a humanoid

Today's dataset ideas, applied to a robot shaped like a person

Extension material: dataset structure, collection methods, quality and coverage, and domain randomisation, on a full humanoid.

A humanoid dataset is still a time series

One episode = one attempt, from reset to finish. Same structure as slide 6, bigger frames.

frame 0	frame 1	frame 2	frame 3	…	frame 599
head image	head image	head image	head image		head image
28 joints	28 joints	28 joints	28 joints		28 joints
28 actions	28 actions	28 actions	28 actions		28 actions

Frame
one time step: observation + action + timestamp

Episode
the frames of one attempt, in order

Dataset
many episodes plus metadata: robot type, fps, task text

Example: a public Unitree G1 dataset on the Hugging Face Hub, recorded at 30 fps in LeRobot format v3.0.

Inside one humanoid frame

Field	Shape	Meaning
observation.state	28 numbers	follower joints: 7 per arm, 7 per hand
observation.images.cam_left_high	480 × 640 × 3	head camera image
action	28 numbers	commanded joints, from the operator's arms and hands
timestamp · frame_index · episode_index	numbers	where this frame sits in time
task	text	for example “pick up the cube”

Often added on humanoids

IMU (body tilt and spin) · walking speed · depth images · more cameras (AgiBot World records 8) · touch sensors in the grippers

Worked example: how big is a humanoid dataset?

Same plan as slide 11 (50 episodes × 20 s × 30 fps), on a humanoid with 28 joints and 3 cameras

30,000
frames (unchanged)

90,000
camera images (3 cameras)

840,000
joint values in the state (was 180,000)

≈ 1–2 h
to record, with resets and VR set-up

At company scale: AgiBot World (2025)

≈ 1,000,000 episodes, **≈ 3,000** hours, **100+** robots, **217** tasks, **106** scenes, **3,000+** objects

Collected by trained teleoperators in a 4,000 m² data-collection facility.

Recording time is an estimate; the other numbers follow from the plan above.

Collecting humanoid data: three ways, plus one

Teleoperation

VR hand tracking, motion-capture suit, exoskeleton or a webcam.

Needs retargeting: a human body is not a robot body.

Used by AgiBot World, Atlas, Open-TeleVision.

Kinesthetic teaching

Hard on a whole humanoid: it is heavy, and it can fall while you push it.

Sometimes used on the arms only, with the robot held safely.

Randomised simulation

The main source for walking and balance.

Thousands of virtual humanoids, with different floors, masses and pushes every episode.

New: human video

People film everyday tasks; no robot needed.

Figure taught its Helix model to walk to places in a home from human video alone.

No robot joints or actions are recorded.

Slide 13's test still works: an exoskeleton is teleoperation, because your hand never touches the robot doing the task.

Teleoperating a humanoid: the device leaves a signature

Device	What it captures	Signature it leaves in the data
VR hand tracking	head, wrists and fingers; the robot's head camera streams to the headset	natural motion; small jitter and delay; no sense of touch
Motion-capture suit or trackers	the whole body, including steps	whole-body moves; needs a large, clear space
Exoskeleton	the operator's joint angles directly	smooth and joint-matched, like our leader arm
One webcam	body and hand pose estimated from video	cheapest; the noisiest and least precise

Slide 14 still holds: the policy copies whatever style the data has.

The data pyramid: how humanoid teams mix their data

Real robot teleoperation
precise, expensive

Simulation
cheap, scales, reality gap

Human video
huge, cheap, no robot actions

Top: most realistic, least of it
AgiBot World: about 3,000 hours from 100+ robots

Middle: cheap to multiply
NVIDIA GR00T: 750,000 simulated demos in 11 hours

Base: the most of it
Figure Go-Big: video from real homes, no robot needed

Slide 19 again: no method wins on everything, so humanoid teams stack all three.

Good demo, bad demo: humanoid edition

Keep

- smooth, direct arm and hand motion
- one consistent strategy, the same hand every time
- the operator keeps looking at the task, so the head camera sees it
- the robot stays steady and balanced

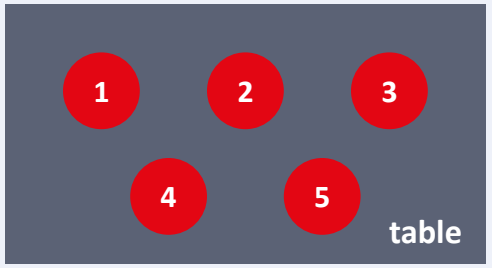
Discard

- a tracking glitch: a hand suddenly jumps
- hesitation from VR delay, or long idle frames
- the operator looks away, so the camera misses the grasp
- a wobble, a stumble or a near-fall

How AgiBot World checks quality

Trial run to set standards → operators check each episode before upload → annotators verify every episode. Idle frames are removed; failed-then-recovered episodes are kept but labelled (about 1%).

Coverage for a robot that walks



cube positions 1–5 on the table
robot start spots A–C (it walks to the table)



5 positions × 3 start spots × 4 demos = 60 episodes

Where the robot starts

a humanoid walks to the task, so vary its start position and direction

Rooms, objects and lighting

AgiBot World: 106 scenes and 3,000+ objects

Operators

different people move differently; mixing operators widens the data

Domain randomisation for a walking robot

Dynamics

floor friction · body mass · motor strength · sudden pushes

e.g. friction 0.5–1.25 · motor strength ±20% · a push of up to 1 m/s every 15 s

Appearance

floors, walls, lighting, textures

matters for camera-based policies; many walking policies use no camera at all

Sensing

noise on joint and IMU readings · delay

e.g. 0–60 ms of action delay, close to the 20–50 ms real robots show

New for humanoids: random pushes

The simulator shoves each virtual robot now and then, so the policy learns to recover its balance before it ever meets a real bump.

Worked example: three randomised humanoid episodes

Randomised term	Range	Episode 1	Episode 2	Episode 3
Floor friction	0.5 – 1.25	0.62	1.10	0.81
Extra body mass	−2 to +2 kg	+0.8 kg	−0.5 kg	+1.6 kg
Motor strength	×0.8 – ×1.2	×0.93	×1.12	×0.85
Push	≤ 1 m/s, every ~15 s	0.7 m/s	0.3 m/s	0.9 m/s
Action delay	0 – 60 ms	20 ms	0 ms	40 ms

Too narrow: it walks only on the simulator's perfect floor.

Tip: get stable walking first, then widen the ranges (slide 26).

Illustrative values inside typical ranges; a real run draws new values for each of thousands of simulated robots.

What you would need to collect humanoid data

Simulation data
start here, with Lab 4's tools

Hardware: an NVIDIA RTX PC like our Isaac Sim machines.

Software: Isaac Lab humanoid robots with domain randomisation; Isaac Lab Mimic to grow a few demos into many; LeRobot's MuJoCo Unitree G1.

Teleoperation data
a realistic lab set-up

Hardware: Unitree G1 (basic model from US$13.5K); Apple Vision Pro, Meta Quest 3 or PICO 4 Ultra; or an exoskeleton.

Software: Unitree xr_teleoperate, or LeRobot's lerobot-record for the G1, saving in LeRobot format.

Human video data
the cheapest to collect

Hardware: a head-mounted or chest camera, or a phone.

Software: pose estimation to find hands and body; the actions still have to be worked out from the video.

The common thread: LeRobot-format datasets, the same structure you inspect in Lab 4.

Discussion: collecting data for a humanoid

1. **Outcome 1 ·** A humanoid dataset: 40 episodes of 15 s at 30 fps, 28 joints, 3 cameras. How many frames, images and state values?
2. **Outcome 2 ·** Why is kinesthetic teaching rare for a whole humanoid, and why is an exoskeleton still teleoperation?
3. **Outcome 3 ·** In VR the robot's head follows yours. Write one keep-or-discard rule that comes from this.
4. **Outcome 4 ·** Name the reality gap that each of these closes: random pushes · IMU noise · random floor colours.

Pairs, 10 minutes. Be ready to share one answer with the class.

References: advanced topic (humanoids)

AgiBot World Colosseo: a large-scale manipulation platform (2025). https://arxiv.org/abs/2503.06669

Figure AI (2025). Project Go-Big: internet-scale humanoid pretraining and direct human-to-robot transfer. https://www.figure.ai/news/project-go-big

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Boston Dynamics (2025). Large behavior models and Atlas find new footing. https://bostondynamics.com/blog/large-behavior-models-atlas-find-new-footing/

Example Unitree G1 dataset in LeRobot format. https://huggingface.co/datasets/Breno-de-Angelo/unitree-g1-manipulation

LeRobot documentation: Unitree G1. https://huggingface.co/docs/lerobot/unitree_g1

Legged Gym Ex: domain randomization tuning guide. https://leggedgym-ex-doc.readthedocs.io/en/latest/developer_guide/how_to/domain_randomization.html